

Cybersecurity Incident: "Syn Flood"

Section 1: Identify the type of attack that may have caused this network interruption

The attack threatening the network's availability is a DoS attack originating from IP address 203.0.113.0. The attacker is using a SYN flood technique to slow down the server, which can be observed through the increased response times and delayed server performance.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors attempt to establish a connection with the web server, a TCP three-way handshake takes place.

1. The client sends a SYN (Synchronize) packet to the server, requesting to establish a connection.
2. Under normal circumstances, the server responds with a SYN-ACK packet, indicating that it received the request and is ready to communicate.
3. The client then sends an ACK packet, completing the handshake and establishing the TCP connection.

In this incident, the attacker abused the first step of the handshake by sending a large number of SYN packets to the server without completing the connection process. This type of attack is known as a SYN Flood attack.

When an actor sends an excessive number of SYN packets, the server allocates resources for each connection request and eventually becomes overwhelmed. As a result, legitimate users may be unable to establish connections with the server, causing a DoS condition.

The logs show a significant increase in response times during the attack. Prior to the attack, response times were approximately 3.144521 and 3.195755 seconds. During the attack, response times increased to 49.561927 and 51.823018 seconds, indicating severe degradation of service performance as the server became saturated.